MUZAMMAL SHAFIQUE

Student Member, IEEE
+1 (810) 875-1565 ◇ muzshf@gmail.com ◇ Michigan, United States
Homepage ◇ Google Scholar ◇ LinkedIn ◇ GitHub

EDUCATION

University of Michigan-Flint **January 2025 - April 2026**
Masters of Science in Computer Science

- **Courses**: Deep Learning, Algorithm & Complexity Analysis, Natural Language Processing
- **Thesis**: "*Towards Reliable Intracranial Aneurysm Analysis: Leveraging Foundation Models for Structure-Aware Detection and Segmentation across 3D CTA, ToF-MRA and DSA*"

National University of Sciences and Technology (NUST) **September 2019 - May 2023**
Bachelors of Science in Mathematics

- **Courses**: Calculus, Mathematical Computing, Probability Theory, Statistics, Linear Algebra
- **Final Year Project**: "*Banach Fixed-Point Theorem in Proximity Spaces*"

EXPERIENCE

General Motors **August 2026 - Present**
Machine Learning Engineer

- Currently working on validating and fixing 3D CAD files of assemblies and sub-assemblies of vehicle as per GM's policy requirement.

NUSTIAN **August 2023 - Present**

Registrations Lead *January 2025 - Present*
Data Analyst Consultant *May 2024 - December 2024*
Data Engineer *August 2023 - April 2024*

University of Michigan-Flint **January 2025 - April 2026**
Graduate Research Assistant

- Worked on:

  1. Artery-Aware Intracranial Aneurysm (IA) Segmentation and Rupture Prediction using VISTA3D foundation model: Achieved dice score of 0.9287 and 0.8415 on 6 class 3D cerebral arteries segmentation on ToF-MRA and CTA respectively
  2. Multimodal Physics-Informed Graph Neural Networks for Intracranial Aneurysm Hemodynamics and Rupture Prediction: Preliminary results show a mean absolute error (MAE) of 0.1721 m/s in velocity magnitude

- Worked on boundary-aware medical image segmentation of diverse imaging modalities including DSA, CTA, MRI and Ultrasound: Fine-tuned MedSAM
- Collaborated directly with a leading neurosurgeon at Henry, Dr. `Ghaus Malik`, at Henry Ford Hospital, gaining deep clinical insight into aneurysm morphology, rupture-critical biomarkers, cerebral artery branching patterns, and artery-specific velocity profiles and vessel diameters

National University of Sciences and Technology (NUST) **June 2023 - July 2023**
Research Assistant

- Built ML models, forecasted monthly fuel consumption for chemical plant chimneys, quantified prediction uncertainty using Bootstrapping and PCE, delivering 95% confidence intervals with more than 90% coverage to enable risk-aware energy planning and cost control

Oeson Learning **March 2023 - May 2023**
Data Science Intern

· Built a movie recommendation engine through unsupervised learning using Streamlit. Learnt different tools and concepts of ast.literal_eval, PorterStemmer, text vectorization using bag of words technique, cosine similarity

PROJECTS

Vision-Language Model Adaptation in Radiology ***Fall 2025***
Fine-tuned LLaMA-3.2-11B Vision-Instruct with LoRA on radiology image-report pairs to generate pathology-aware clinical narratives.

RAG-Based Medical Assistant for Brain Aneurysm ***Summer 2025***
Developed a retrieval-augmented generation (RAG) medical assistant grounding aneurysm-related queries in clinical evidence.

PUBLICATIONS

1. **Muzammal S.** et al. "**ARAN**: Leveraging Foundation Models for Vasculature-Tree-Informed **AR**tery-Aware Intracranial **AN**eurysm Detection in 3D CTA and ToF-MRA", *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026*
2. **Muzammal S.** et al. "**FocusSDF**: Boundary-Aware Learning for Medical Image Segmentation via Signed Distance Supervision", *Proceedings of the 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI), (London, UK)*
3. Nasir R., **Muzammal S.** et al. "**CrossStream-Seg**: A Cross-Guided Two-Stream Learning for Boundary-Sensitive ROI Segmentation", *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026*

TECHNICAL SKILLS A& EXPERTISE

Scientific Computing	Python, R, C/C++, MATLAB, Data Processing, Model Evaluation
AI, Deep Learning, Vision	PyTorch, TensorFlow, nnUNet, MONAI
Foundation Models, GenAI	VISTA3D, MedSAM, Multimodal Llama, RAG, AI Assistants
HPC, & Research Tools	Large Scale GPU Computing, DDP, Slurm, Turbo Storage, Globus
Medical Imaging Modalities	DSA, CTA, MRA, Ultrasound, X-Ray, Axial, Coronal, Sagittal views, DICOM, NIfTI

RESEARCH INTERESTS

3D Medical Image Analysis, Computer Vision, Vision Foundation Models, VLMs and LLMs in medical imaging, Multimodal Learning, Cerebrovascular and Cardiovascular Blood Hemodyanmics Prediction

CERTIFICATIONS

· Machine Learning Specialization - **Stanford University**
· Data Science - **Harvard University**
· Generative AI - **iNeuron**

AWARDS & HONORS

· Awarded a fully sponsored trip, including paper registration, to attend CVPR 2026 (June) to present my paper.
· University of Michigan Conference Travel Award of $5000 to attend ISBI 2026 in London, UK
· Awarded $1000 in "AI in Research Symposium 2026", University of Michigan's one of most prestigious award
· University of Michigan Non-Resident Graduate Scholarship
· Prime Minister's Ehsaas Undergraduate Scholarship (fully funded award for complete BS degree)
· Dean's Merit Scholarship Award (NUST)
· Offered Russian Government Scholarship 2023 for Masters in Artificial Intelligence